Run report: run-20261011T1955Z-verify

Eigen-Séta v2: correctness rates, constant-time probes at Toy16 and L1, timing test, filter rates (descriptive record)

Study: research/eigen_seta_rust_20261011 · sources: commit d504af28e (filter step: 72051ac67)

What ran

Times (UTC)	Start 19:56:36Z, end of phase 1 20:29:49Z, end 20:37:48Z (2026-10-11). Filter step 20:38:30Z–20:46:52Z.
Sources	Commit d504af28e with a clean worktree. The filter step used 72051ac67, which differs only by capping the public basis search and adding the filter example.
Build / tools	rustc 1.97.0, release with debuginfo. valgrind-3.22.0 (callgrind), glibc malloc.
Host	Intel Xeon @ 2.80 GHz, 4 logical cores, Linux 6.18.44 x86_64, shared cloud VM
Phase 1	4 callgrind probes at L1 in parallel with native failure-rate runs: Toy16 ×3000, Toy32 ×600, L1 ×20
Phase 2	12-run Toy16 probe grid: keys 1–4 × (valid 1, valid 2, foreign)
Phase 3	`dudect_decaps 1000`, alone on the machine
Probe	`valgrind --tool=callgrind --toggle-collect='*probe_core*' ./ct_probe <key> <ct> <mode> <set>`. Only `decaps_core` is measured. Allocator = glibc malloc and arena code, sbrk, brk, morecore, mem*.
Records	`records/` (failrate/, toyprobe/, l1probe/ with callgrind/*.xz, dudect_output.txt, f_filter_rate.txt, verify_run.sh, ctsummary.sh, build.log, commit.txt)

Requested and completed: 4 + 12 probes, 3,620 trials, 1 timing test, 5 filter measurements. Failures: none.

Decapsulation correctness

Set	Trials	Correct	Failures
Toy16	3,000	3,000	0
Toy32	600	600	0
L1	20	20	0

Instructions in `decaps_core`

set	key	ct	acc.	total	allocator	rest
Toy16	1	valid 1	yes	1,147,950,979	31,741,763	1,116,209,216
Toy16	1	valid 2	yes	1,147,950,979	31,741,763	1,116,209,216
Toy16	1	foreign	no	1,148,059,231	31,850,015	1,116,209,216
Toy16	2	valid 1	yes	1,147,388,400	31,179,184	1,116,209,216
Toy16	2	valid 2	yes	1,147,388,400	31,179,184	1,116,209,216
Toy16	2	foreign	no	1,147,693,051	31,483,835	1,116,209,216
Toy16	3	valid 1	yes	1,147,459,087	31,249,871	1,116,209,216
Toy16	3	valid 2	yes	1,147,459,087	31,249,871	1,116,209,216
Toy16	3	foreign	no	1,147,849,768	31,640,552	1,116,209,216
Toy16	4	valid 1	yes	1,147,477,368	31,268,152	1,116,209,216
Toy16	4	valid 2	yes	1,147,477,368	31,268,152	1,116,209,216
Toy16	4	foreign	no	1,147,787,126	31,577,910	1,116,209,216
L1	1	valid 1	yes	124,250,795,358	3,254,365,748	120,996,429,610
L1	1	valid 2	yes	124,250,795,358	3,254,365,748	120,996,429,610
L1	1	foreign	no	124,257,288,150	3,260,858,540	120,996,429,610
L1	2	valid 1	yes	124,262,712,389	3,266,282,779	120,996,429,610

Observation: at each set the rest takes a single value. The allocator count varies with key and mode, and is equal for the two valid ciphertexts of a key. A first L1 summary that did not count sbrk and brk as allocator code showed a 174-instruction difference between the keys, all of it in those libc heap-growth routines.

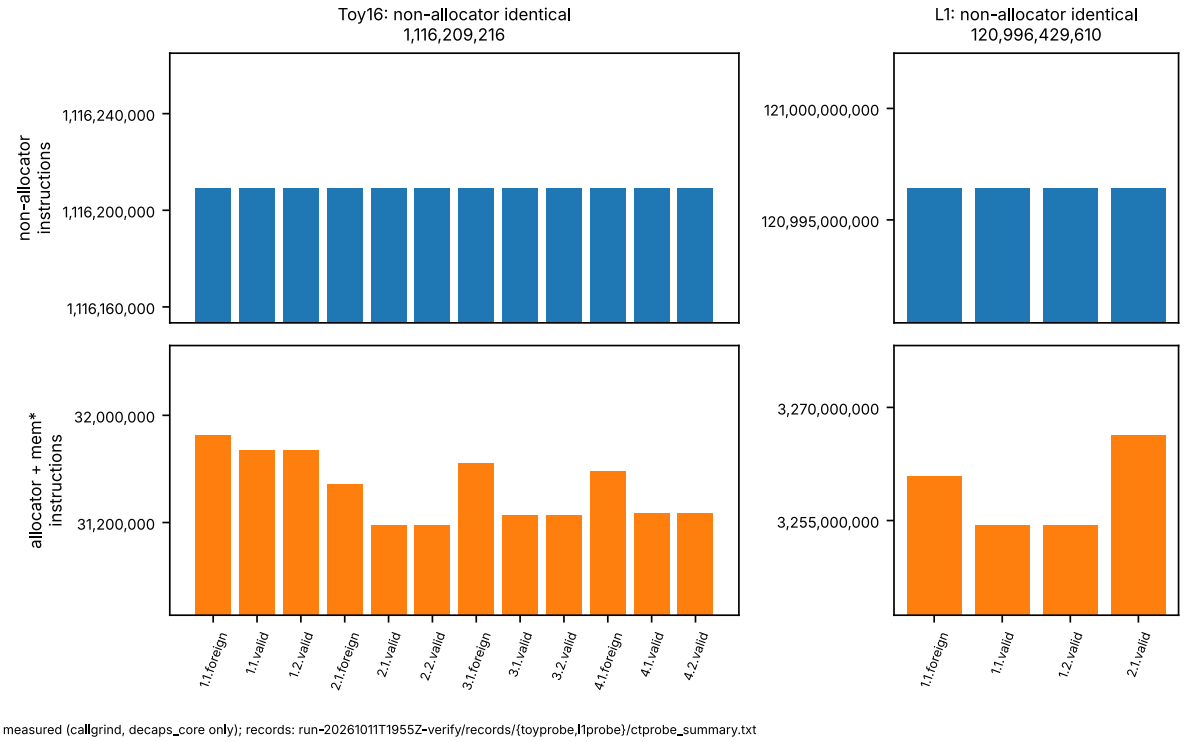


Figure 1: Instruction counts per run, Toy16 and L1. Measured; records/{toyprobe,l1probe}/ctprobe_summary.txt.

Timing test (ns per call, 1,000 samples per class)

```
samples/class 1000; median ns: fixed 122373850, random 120844155
t(raw) = +1.17
t(crop <= p50) = +0.14  (n = 473, 528)
t(crop <= p75) = +1.46  (n = 730, 771)
t(crop <= p90) = +2.36  (n = 899, 902)
```

Filter acceptance rates (fraction of odd f candidates)

Set	Drawn	Accepted	Rate	log2
Toy16	12,408	300	0.02418	−5.37
Toy32	24,956	300	0.01202	−6.38
L1	55,933	200	0.00358	−8.13
L3	39,987	100	0.00250	−8.64
L5	51,887	100	0.00193	−9.02

Rejections: gcd(f,21) ≠ 1 (about 43%) and the two-squares step (52–56% of all drawn). The schedule condition rejects about half of the candidates that reach it.

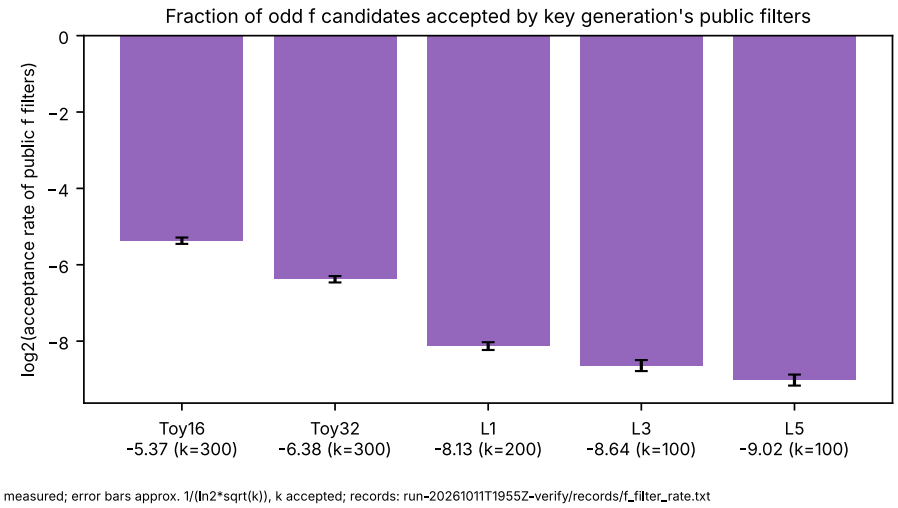


Figure 2: log2 of the acceptance rate; error bars are approx. $1/(\ln 2 \cdot \sqrt{k})$. Measured; records/f_filter_rate.txt.

Scope

Toy16 and L1 probes only, not L3 or L5. One compiler and one target. Instruction counts, not memory addresses. One key in the timing test. Zero failures in 3,000 trials bounds the Toy16 failure rate only by about 10^{-3} at 95% confidence.